

Resting-State EEG Classification of Children and Adolescents Diagnosed with Obsessive Compulsive Disorder Using Convolutional Neural Network

Amir Jahanian-Najafabadi^{1*}, Khaled Bagh^{1,2}, Bernhard Hommel³

^{1*} Department of Cognitive Neuroscience, Faculty of Biology, University of Bielefeld, Bielefeld, 33501, Germany

² Faculty of Technology, University of Bielefeld, Bielefeld, 33501, Germany

³ School of Psychology, Shandong Normal University, Jinan, 250061, China

*Corresponding Authors: Amir Jahanian Najafabadi amir.jahanian@uni-bielefeld.de

Abstract

Obsessive-compulsive disorder (OCD) is a complex psychiatric condition that varies widely from person to person, making early and accurate diagnosis, especially in children and adolescents' challenging. In this study, we applied deep learning techniques, specifically convolutional neural networks (CNNs), to detect neuro-biomarkers associated with OCD in age-matched children and adolescents, comparing those diagnosed with OCD to healthy controls. The models were designed to identify meaningful patterns in brain activity across different frequency bands and regions of interest. Our approach was based on the VGG16 architecture and applied following extensive data preprocessing, which included brain connectivity estimation using a multivariate autoregressive model and independent component analysis. Results revealed that the VGG16 deep learning model performed very well in identifying children and adolescents with OCD using resting-state EEG data. Using the fPPDC connectivity measure, the model achieved an accuracy of 86.6%, a strong F1-score of 85.3%, and high specificity of 94.6%, indicating its ability to reliably distinguish individuals with OCD from healthy controls. Additional performance metrics including a high AUC of 94.8% and an MCC of 76.0% further confirm the model's effectiveness in capturing meaningful brain activity patterns related to OCD. We found stronger resting state functional connectivity within specific brain networks, particularly in the delta, beta, and gamma frequency bands, spanning the frontal, temporal, and occipital regions, and higher beta and gamma activity in the fronto-temporal and occipito-frontal connections. These findings contribute to the growing understanding of OCD's neurobiological basis and help pave the way for EEG-based diagnostic tools tailored to younger populations, particularly children and adolescents.

Keywords: Obsessive-compulsive disorder, Resting-State Electroencephalography, Children, Adolescents, Convolutional Neural Network

Introduction

Obsessive Compulsive Disorder (OCD) is a complex heterogenous psychiatric disorder affecting 1% to 3% of the global population (Ruscio et al., 2010). OCD is marked by intrusive thoughts and repetitive behaviors that can seriously disrupt a person's daily life as well as affecting their family relationships (Pato et al., 2024). Despite the availability of well-established treatments like exposure and response prevention and selective serotonin reuptake inhibitors, up to half of all individuals with OCD don't experience full remission. Several factors complicate effective treatment ranging from misdiagnosis and the wide variety of symptom profiles to family accommodation and frequent co-occurring conditions like depression and anxiety (Reynolds et al., 2023; Clark et al., 2017). Given that current treatment guidelines offer little direction on who is most likely to benefit from specific therapies or who might not respond at all, there's a growing need for more precise diagnostic tools (Krzyszczuk et al., 2019; Alowais et al., 2023).

OCD was previously grouped under anxiety disorders, and its symptoms often overlap with anxiety (Goodwin, 2015). In many cases, people with anxiety may also experience obsessions and compulsions (Leonard & Reimann, 2012). This overlap can make it difficult to diagnose OCD accurately and may lead to delays in treatment. Because of this, identifying reliable biomarkers could help distinguish OCD from other conditions and support more accurate diagnosis. Supporting that, computational methods that can analyze and distinguish complex patient data may offer a path forward, helping clinicians make more informed decisions and paving the way for more personalized approaches to care (Fullana et al., 2020; Li et al., 2023; Kalmady et al., 2022). Thus, identifying OCD early is crucial and it leads to more effective treatment, helps ease symptoms, and improves patient outcomes before it becomes chronic (Fontenelle et al., 2022).

Understanding the biological markers linked to OCD is an important step in diagnosis and can support clinicians as an additional tool in both research and practice. Biomarkers offer objective insight into biological and therapeutic processes (Bandelow et al., 2016). Among these, EEG-based biomarkers are especially promising, as EEG provides a non-invasive way to detect OCD with less reliance on subjective interpretation (Provenza et al., 2022; Zaboski et al., 2021; 2025; Jahanian Najafabadi & Bagh, 2025). Proper EEG analysis can guide accurate diagnosis and support personalized treatment plans. However, relying solely on visual inspection or behavioral assessments and subjective self-report associated with that aren't enough for psychiatric conditions like OCD (Huidobro et al., 2025; Zaboski et al., 2021), since EEG data is complex and full of subtle patterns. To make sense of this data, previous researchers including us used a range of techniques, from traditional mathematical methods to machine learning approaches like principal component analysis (PCA), genetic algorithms, support vector machines (SVM), and linear discriminant analysis (LDA) (Mahato et al., 2019; Bhaskarachary 2020, 2021, Ciodaro et al., 2020; Jahanian Najafabadi & Bagh, 2023, 2024, Bagh & Jahanian Najafabadi, 2024).

More recently, deep learning has shown strong potential in this area (Askland et al., 2015). A number of studies have looked into how effective and accurate neural networks (NNs) are when

it comes to analyzing EEG data. For example, Roy et al. (2019) reviewed over 150 studies and found that a major trend in this field is using NNs for classification tasks involving EEG signals. Among the most commonly used NN models were CNNs, recurrent neural networks (RNNs), autoencoders (AEs), and Long-Short-term Memory (LSTM). Overall, these neural network approaches showed about a 5.4% improvement in performance compared to more traditional methods and benchmarks. In an earlier study, Altuğlu et al (2020) used machine learning with EEG band power to predict how well patients with OCD would respond to transcranial magnetic stimulation, and the theta band alone gave us about 80% accuracy. They further applied machine learning to identify treatment-resistant OCD using EEG complexity measures and achieved 90% accuracy.

Moreover, a recent study by Ahmed et al. (2024) revealed 96.83% accuracy in detecting OCD using artificial neural network (ANN) while Basak et al. (2023) reported detection of OCD with 95% precision using LSTM. However, the number of datasets and channels (19 vs 64) were inconsistent. Another study by Farhad et al. (2024) demonstrated high accuracy during both cross-validation and external testing. The one-dimensional CNNs-LSTM model achieved an average accuracy of about 90.9%, while the one-dimensional CNNs –Gradient Recurrent Units’ model followed closely with around 85.9%. The most informative signals came from the inferior frontal, temporal, and occipital brain regions highlighting the potential of combining deep learning methods with EEG data to reliably distinguish between individuals with OCD and healthy controls. Verma et al. (2024) further reported beta band functional connectivity achieved 90% accuracy in OCD patients, outperforming the 74.52% accuracy using Gamma band reported by Park et al. (2021).

To our knowledge, even though there have been several studies in this area and particularly only among adults (see Ma et al., 2025; Mukherjee et al., 2024; Ren et al., 2024; Park et al., 2021), there still isn’t a clear, well conceptualized and designed study or a promising optimized model that can optimally detect abnormal biomarkers to distinguish OCD from healthy individuals, especially focusing on children and adolescents. The research so far has been inconsistent, particularly when it comes to factors like participant age, gender and sample size. These gaps led us to explore the use of CNNs to classify both normal and abnormal biomarkers in psychiatric conditions, with a focus explicitly on OCD in children and adolescents' populations, hoping to contribute to ongoing research and future clinical applications.

Methods and materials

For this study, we analyzed resting-state EEG data from 315 American children and adolescents aged 5 to 21. Of these, 39 had been diagnosed with OCD, while the remaining 276 were not given any diagnosis, classified as healthy and labeled as HBN by Child Mind Institute. The data came from the publicly available Healthy Brain Network (HBN) dataset (Alexander et.al., 2017). The

recordings were taken using a 128-channel EEG (cf., Table 1) HydroCel Geodesic system (Electrical Geodesics Inc.), with a sampling rate of 500 Hz and a frequency range of 0.1 to 100 Hz. However, after removing the outer channels, only 109 channels were kept for analysis (cf., Table 2). The arrangement of EEG electrodes on the scalp is shown in Figure 1 (A), divided based on Regions of Interest (ROI, cf., Figure 1 B), and a summary of the dataset is depicted in Table 3.

Table 1. 128 Channels grouped into 7 clusters based on the 7 regions of the brain.

Region	Channels
Frontal Right Lobe	E1, E2, E3, E4, E5, E8, E9, E10, E14, E118, E123, E124
Frontal Left Lobe	E11, E12, E15, E16, E17, E18, E19, E20, E21, E22, E23, E24, E25, E26, E27, E32
Temporal Right Lobe	E86, E92, E93, E97, E98, E100, E101, E102, E103, E104, E108, E109, E110, E111, E114, E115, E116, E117, E121, E122
Temporal Left Lobe	E28, E29, E33, E34, E35, E36, E38, E39, E40, E41, E42, E44, E45, E46, E47, E50, E51, E52, E53, E57
Center	E5, E6, E13, E30, E31, E37, E54, E55, E79, E80, E87, E105, E106, E112
Occipital Right Lobe	E76, E77, E78, E82, E83, E84, E85, E89, E90, E91, E95, E96
Occipital Left Lobe	E58, E59, E60, E61, E62, E64, E65, E66, E67, E69, E70, E71, E72, E74, E75

Table 2. Excluded outer channels

Outer Channels
E43, E48, E49, E56, E63, E68, E73, E81, E88, E94, E99, E107, E113, E119, E120, E125, E126, E127, E128

Table 3. The healthy brain network dataset parameters

Parameter	Description
Subjects	315 participants (39 OCD, 276 HBN)
Channels	128 raw channels + Cz (109 after pre-processing)
Sampling rate	500 Hz
Avg. Dataset Recording Duration	400 seconds (about 6.7 minutes)
Pre-processing	Illustrated in figure 2
Dropped Datasets (after pre-processing)	7 (7 HBN), after pre-processing
Selected Datasets (for the Neural Networks)	76 (38 OCD and 38 HBN)

The demographic data of the participants were also provided by the HBN Institute, as shown in Table 4. The average age of participants in the healthy dataset (labeled as HBN) was 11.0 ± 4.11 years, while the average age of participants with OCD was 11.8 ± 3.92 years. Although other phenotypic data such as information on medications and results from the Child Depression Index surveys were available, most participants chose not to complete them. Therefore, these data are not included in this study.

Table 4. Demographic information across healthy (HBN) and OCD, showing the number of samples per group, and showing the mean and standard deviation of the age group.

Class	Children	Adolescent	Male	Female
HBN	26 (8.4 ± 1.99)	12 (13.8 ± 0.86)	25	13
OCD	24 (8.8 ± 1.99)	14 (14.5 ± 1.59)	20	18

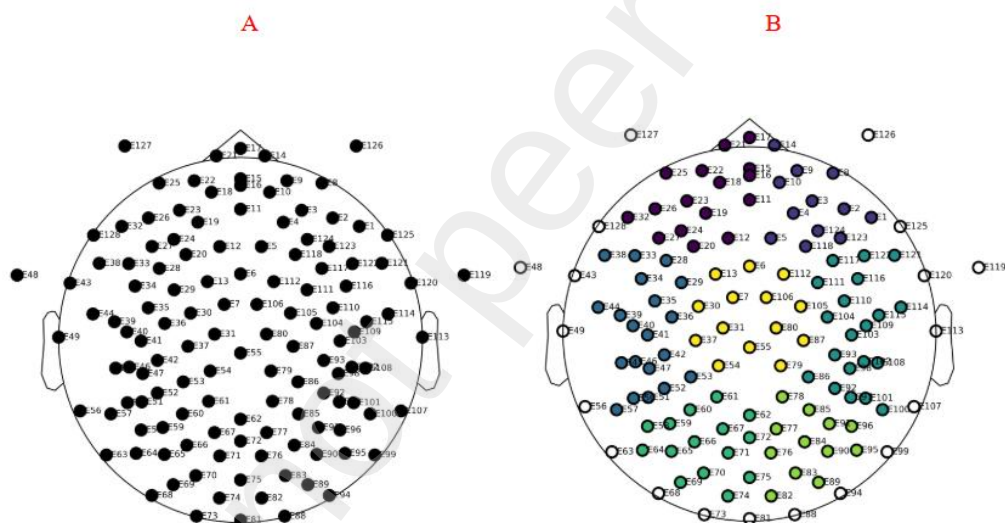


Figure 1. A. Channel location of 128-channels in the HBN dataset; B. Regions of Interest (ROI)

Pre-processing

The data pre-processing was divided into two main stages: (1) EEG data pre-processing and (2) computation of the connectivity model and connectivity measurements. The EEG pre-processing steps are illustrated in Figure 2. The process begins with loading the raw EEG data using the MNE library in Python, applying the GSN-HydroCel-128 montage. Outer channels and any zeroed channels resulting from poor recording quality or filtering during data acquisition were removed. A zero-phase band-pass filter is then applied, with a low-pass cutoff at 70 Hz and a high-pass cutoff at 1 Hz. Additionally, a notch filter at 60 Hz is used to eliminate power line noise, which is common in data recorded in the USA. The data is subsequently resampled to 256 Hz.

To detect bad channels, we employed the PyPREP pipeline, which integrates multiple methods for identifying problematic channels, including:

- **Missing Values:** Channels with missing values.
- **Flat Lines:** Channels with no signal variation.
- **Deviation:** Channels with unusually high or low deviations.
- **High-Frequency Noise:** Channels with excessive high-frequency activity.
- **Correlation:** Channels with low correlation to other channels, indicating poor contact or malfunction.
- **Signal-to-Noise Ratio (SNR):** Channels with low SNR.
- **Dropouts:** Channels with intermittent signal loss. i.e. channels that frequently lose signal are marked as bad.
- **Random Sample Consensus (RANSAC):** RANSAC is a robust method for detecting outliers. It identifies channels that consistently deviate from the expected signal pattern based on a consensus of other channels.

The PyPREP pipeline is run with three iterations of RANSAC sample-wise. Detected bad channels are then interpolated and re-referenced based on the average of the remaining channels.

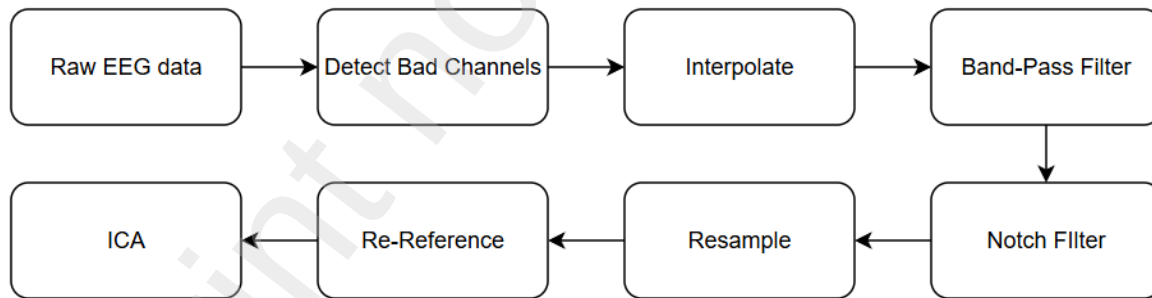


Figure 2. Flowchart of the pre-processing and cleaning of EEG data

Finally, independent component analysis (ICA) is applied using the FastICA model, with the number of components computed as show in the equation below

$$Components_n = Channels_m - Channels_{bad} - 1$$

Where $Components_n$ is the number of possible components, $Channels_m$ is the number of channels and $Channels_{bad}$ is the number of bad channels that could not be interpolated. The ICA model is then fitted to the EEG data and used to identify and remove bad artifacts.

Resting-state Functional Connectivity

After preprocessing and cleaning the EEG data, we computed the resting-state functional connectivity (rsFC) using a Multivariate Auto-Regressive Model with Independent Component Analysis (MVARICA). Two key parameters of the MVARICA model were optimized: the first was the model order (also known as lag), which determines how many previous data points the model uses to predict the current one. The second parameter was the delta ridge penalty, a regularization term used to stabilize the estimates.

For our analysis, rsFC was computed on segmented EEG data, with each segment lasting 15 seconds. However, parameter optimization was performed per participant (rather than per segment), as MVARICA requires a sufficient amount of data to avoid overfitting. Although the data were divided into 15-second segments, optimizing the model per segment would have required fitting the model on 1-second epochs within each segment. Using a single model order per participant also helps reduce spurious variability. This is particularly important because OCD data often exhibit higher noise levels such as from movement or drowsiness, and frequent state shifts. These artifacts are common confounding factors that may lead to false group differences, rather than reflecting true biomarkers of rsFC.

To optimize the model order, we searched within a range of 1 to 30, which is considered suitable for resting-state EEG analysis (Ding et.al., 2000). For each candidate model order within this range, a separate MVARICA model was fitted to the epochs, and the Akaike Information Criterion (AIC) score was calculated. The model order that yielded the lowest AIC score was selected as the optimal value for this parameter. The AIC score is computed using the equation shown below:

$$AIC = 2k - 2 \ln(L)$$

Where k is the number of model parameters, $\ln(L)$ is the log-likelihood of the model. Where the log-likelihood is computed in the equation below:

$$\ln(L) = -\frac{N}{2}(\ln(2\pi\sigma^2) + 1)$$

Where N is the number of samples, σ^2 is the residual variance, and the residual variance is computed as:

$$\sigma^2 = \frac{1}{N \cdot C} \sum_{i=1}^N \sum_{j=1}^C E_{i,j}^2$$

Where C is the number of EEG channels and E is the residual at time i and channel j .

As for the delta ridge parameter, it was optimized using the bisection search method (van de Wiel et.al., 2021).

After fitting the MVARICA model to participants' data, some models may be deemed unreliable due to factors such as excessive noise (e.g., motion artifacts), non-stationarity, or suboptimal model order selection. To systematically identify and exclude poorly fitted models, we implemented a two-stage filtering approach based on residual variance and AIC scores. In the first stage, models with residual variance values greater than 1 were excluded. In the second stage, we identified and removed AIC outliers within each diagnostic group (HBN and OCD) using the Median Absolute Deviation (MAD) method. Together, both filtering stages resulted in the exclusion of 20 datasets 19 from the HBN group and 1 from the OCD group.

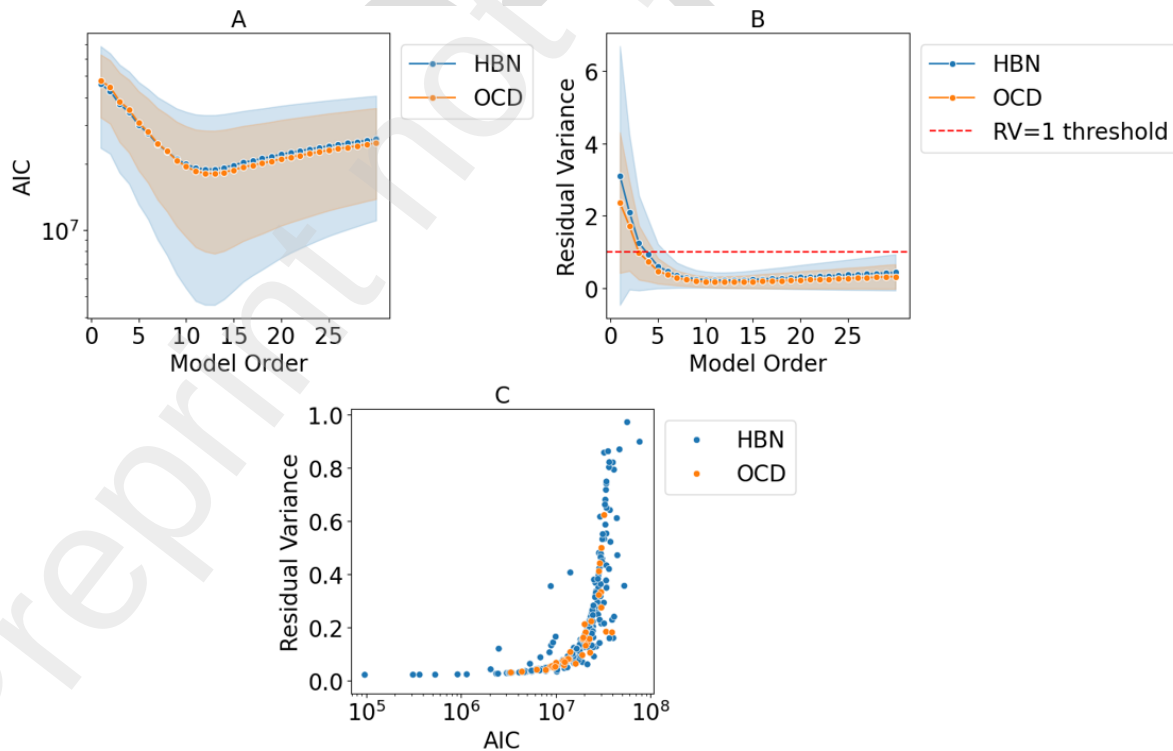


Figure 3. A. Mean and standard deviation of the AIC score across model orders for each diagnostic group. **B.** Mean and standard deviation of residual variance across model orders for each diagnostic group. **C.** Scatter plot of residual variance versus AIC score per participant at their optimal model order, shown separately for each diagnostic group.

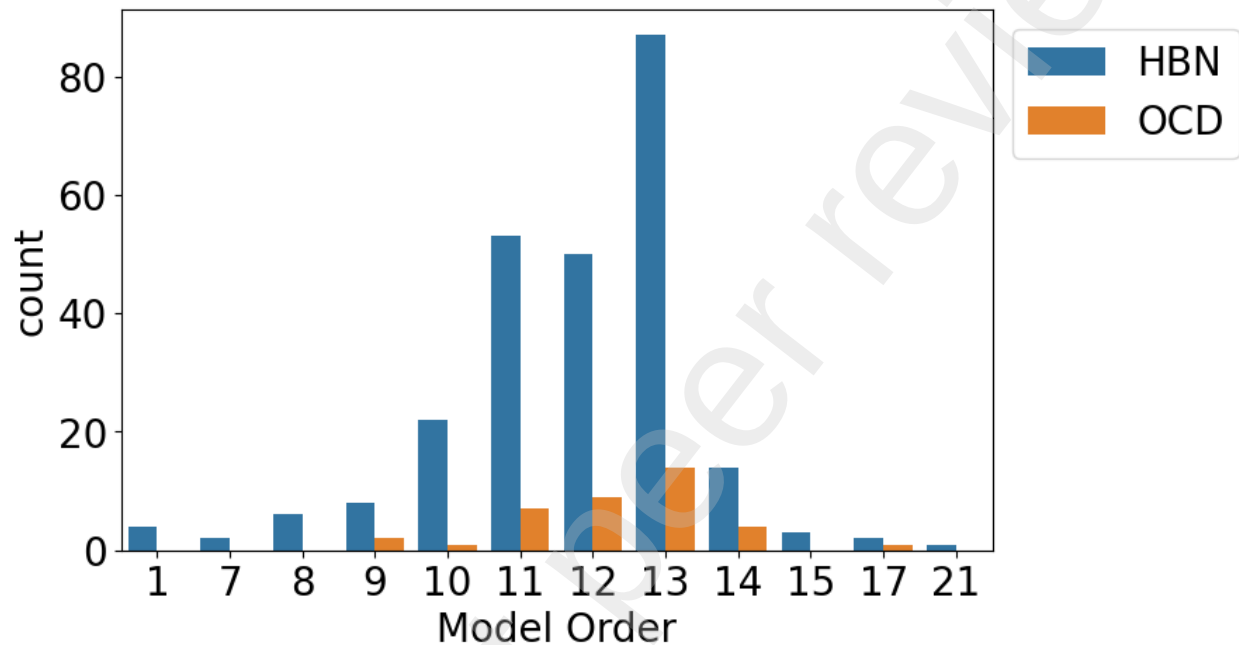


Figure 4. Histogram of model order selection, per diagnostic group.

For each 15-second epoch, once the MVARICA model was optimized, it was fitted to the original EEG segments (Gómez-Herrero et.al., 2008), and various connectivity measures were computed from the fitted model (Faes et.al., 2012). These measures included Partial Directed Coherence (PDC), full-frequency Partial Directed Coherence (ffPDC), and Partial Directed Coherence Factor (PDCF), all calculated bidirectionally between every pair of EEG channels. Connectivity was computed with a frequency resolution of 0.1024 Hz (i.e., 2,500 points), covering the full range from 0 Hz to the Nyquist frequency (0–128 Hz), as shown in Figure 5A.

The results were then divided into standard EEG frequency bands: Delta, Theta, Alpha, Beta, and Gamma. Each connectivity measure within a frequency band was used to generate a matrix image (Figure 5B), where rows and columns represent EEG channels, and each matrix entry indicates the strength of connectivity between the corresponding pair of channels. Channels were organized by regions of interest (ROIs): FL, FR, TL, TR, OL, OR, and Center. Finally, for each

participant, the connectivity matrices were normalized across all segments using min–max normalization.

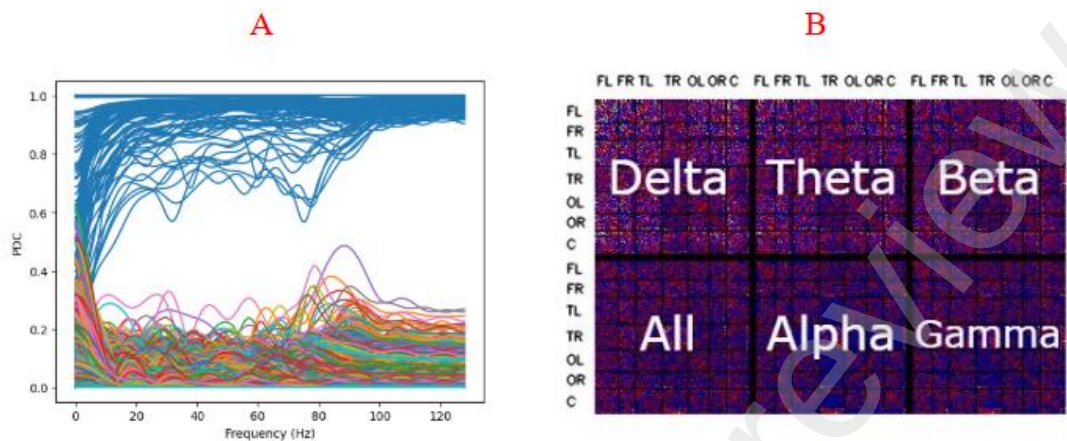


Figure 5. A. Example of computed PDC connectivity for a single segment. **B.** Example of processed connectivity measure images, which is used as input to the CNN modules.

Architecture of CNN

The CNN models utilized here are based on VGG16 and a new suggested model, based on 3d convolution (referred to as CNN-3D). The VGG16 model is pretrained on ImageNet dataset, the model setup and hyper-parameter are based on our previous work (Jahani Najafabadi & Bagh, 2024), which is a basic VGG16 model, similar to the architecture illustrated in Figure 4, with top layers removed (up to block L.11). The input of the model is of shape (218, 327, 3), with the image duplicated across the three channels. The top layers are replaced with a flattening layer, followed by a batch normalization layer, then two fully connected layers with 512 units and ReLU used as activation function. Then a dropout layer of 0.2 was added, followed by another fully connected layer of 512 units, finally, a decision layer with sigmoid as activation function.

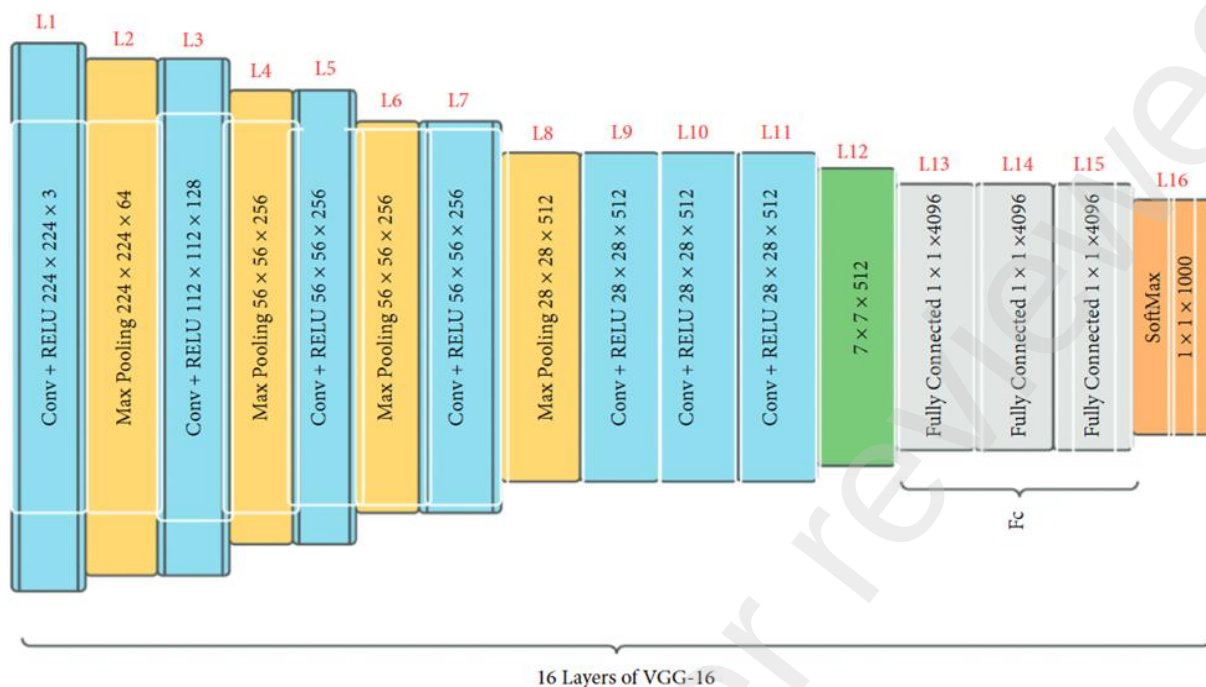


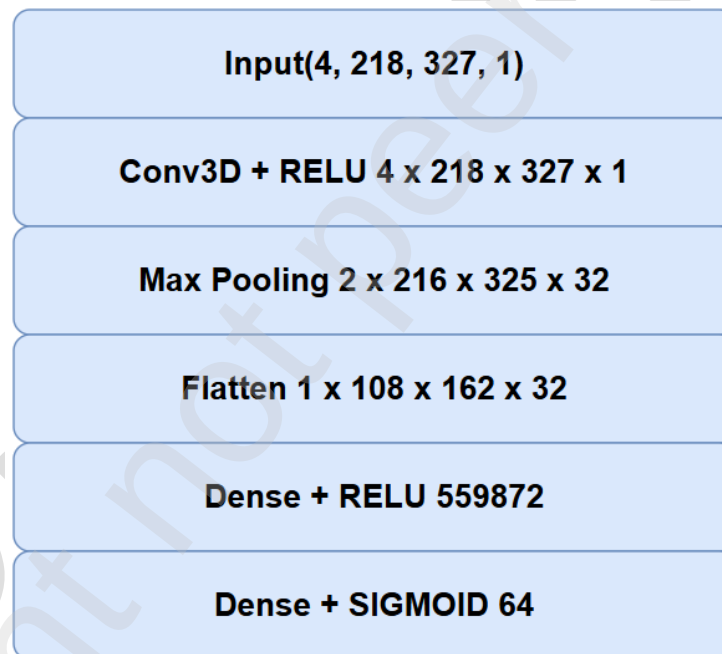
Figure 6. Convolutional Neural Networks' (CNNs) VGG16 architecture (Hasan et.al., 2021)

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258 For the CNN-3D model, the input consists of four temporal EEG segments with a single channel,
259 resulting in an input shape of $4 \times 218 \times 327 \times 1$. The architecture of the model is relatively simple, as
260 illustrated in Figure 5. It begins with a single 3D convolutional layer consisting of 32 filters and a
261 kernel size of $3 \times 3 \times 3$, using the ReLU activation function. This is followed by a 3D max pooling
262 layer, then a flattening layer. The flattened output is passed to a fully connected (dense) layer with
263 64 neurons and a ReLU activation function. Finally, the model includes an output layer with a
264 single neuron and a sigmoid activation function for binary classification.

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268 Figure 7. Convolutional Neural Networks' (CNNs) architecture with 3d convolution layer.

269 ***ROI analysis***

270 When it comes to extracting features importance from the models, a method that focuses on global
271 feature and another that focuses on instance-level explanations were utilized, mainly Permutation
272 Feature Importance (PFI) (Altmann at.al., 2010) and Integrated Gradient (IG) (Sundararajan at.al.,
273 2017).

PFI quantifies the contribution of each feature in a trained machine learning model by assessing the impact of feature perturbation on model performance. In this context, the values of a single ROI interaction are randomly shuffled while all other features remain unchanged. The resulting decrease in model performance is measured using the F1-score which indicates the importance of the shuffled feature. This procedure is repeated ten times to ensure robustness and reliability in the estimation of feature importance.

IG is a popular explainability method, which was used in this work to interpret the predictions of a CNNs model on the test data. The method computes the gradients of the model's predictions with respect to interpolated inputs between a baseline (zero tensor here) and the actual input. Where the integrated gradient is defined as:

$$IG_i(x) = (x_i - x'_i) \frac{1}{m} \sum_{k=1}^m \frac{\partial F \left(x' + \frac{k}{m} (x - x') \right)}{\partial x_i}$$

Where IG is the integrated gradient of input i , x_i is the value of input i , x'_i is the value of the input i in the baseline input, $F(.)$ is the model output function, m is the number of steps used to approximate the integral (50 steps). The gradients are averaged across interpolation steps and multiplied by the input to produce attribution maps, which quantify the contribution of each input feature to the model's output. The results are aggregated across test data used to produce a single score, that expresses the average importance of a single pixel (connectivity channel interaction). This is then averaged across channels to get a score for each ROI, in each frequency band. It should be noted, that for CNN-3D, the integral is computed per time step, per channel interaction.

Both techniques provide insights into the model's decision-making process, with PFI focusing on global feature rankings (e.g., for feature selection), which is also a model agnostic method. Meanwhile, Integrated Gradients quantify the contribution of specific channels to the model's predictions. The difference between IG and PFI, in terms of the meaning of highlighted features, stems from their methodological principles. IG measures how sensitive the model is to changes in a specific feature, indicating how much predictions shift when a feature's value is modified. This can reveal model dependencies on features that may not be causally important but are strongly correlated with the outcome. In contrast, PFI assesses global feature importance by measuring how much performance drops when a feature is randomized. This can overlook redundant features if others compensate. As such, the two methods complement each other—Integrated Gradients show how the model uses features, while PFI reveals which features are statistically critical to performance.

Setup

To ensure a balanced training set, we selected from the HBN class 38 participant dataset (same as OCD), where the criteria for the selection was based on the minimum residual variance value computed. The segmented training dataset was 1865 samples (OCD + HBN) for VGG16 and 441 samples (OCD + HBN) for the CNN-3D, for each connectivity measurement. The training method was 5 K-fold cross-validation training. The fold split was based on subjects. Five different models were trained for each fold. The training was set to 100 epochs with early stopping triggered when the F1-Score hit 0.83, batch size was set to 90, and the learning rate was set to 0.00004. The average training time per epoch is 5 seconds for VGG16 and 1 second for CNN-3D. Additionally, the training of CNNs was done in Python 3.10.12, Tensorflow 2.18.0, Keras 2.18.0, scipy 1.10.1. For each model, the metrics used were accuracy, f1-score, recall, precision, specificity, Cohen Kappa (CK) and Metthew Correlation Coefficient (MCC). Also, the AUC score is used, alongside the Receiver operating characteristic (ROC) curve used to compare the best performing model between VGG16 and CNN-3D.

Results

Table 5 shows the testing score for each CNN module. We observe that in the VGG16 model, the model performed best with FFPDC measurement, achieving a score of 0.866 accuracy, 0.853 f1-score, where recall was 0.795 and precision was 0.814. While specificity was 0.946. Also, CK and MCC were 0.744 and 0.760 respectively. This indicates a solid classification power for the VGG16. With the strongest image classification in the FFPDC measurement, which also aligns with our previous findings (Jahanian Najafabadi & Bagh, 2024). On the other hand, CNN-3D perform strongest with the PDC measurement, achieving an accuracy of 0.798, a f1-score of 0.873 where recall was 0.998 and 0.785, where specificity was 0.636, indicating a bias toward classifying OCD. The CK and MCC scores are 0.646 and 0.700.

Table 5. Testing scores for the best performing model each CNN model type, for each rsFC measurement.

Metric	VGG16			CNN-3D		
	PDC	FFPDC	PDCF	PDC	FFPDC	PDCF
Loss	0.615	0.458	0.856	0.508	0.665	0.701
Accuracy	0.729	0.866	0.734	0.798	0.671	0.568
F1-Score	0.717	0.853	0.729	0.873	0.680	0.619
Recall	0.655	0.795	0.761	0.998	0.771	0.739
Precision	0.814	0.942	0.703	0.785	0.617	0.537
Specificity	0.826	0.946	0.624	0.636	0.542	0.400
CK	0.463	0.744	0.386	0.646	0.312	0.133
MCC	0.482	0.760	0.391	0.700	0.327	0.149

When comparing the two best-performing models, VGG16 and CNN-3D, we observe that although VGG16 has a slightly lower F1-score, it achieves higher overall accuracy and maintains strong specificity. This performance difference is further illustrated in Figure 8, which displays

the Receiver Operating Characteristic (ROC) curves for both models. The Area Under the Curve (AUC) scores are 0.948 for VGG16 and 0.902 for CNN-3D, indicating superior discriminative ability in the VGG16 model.

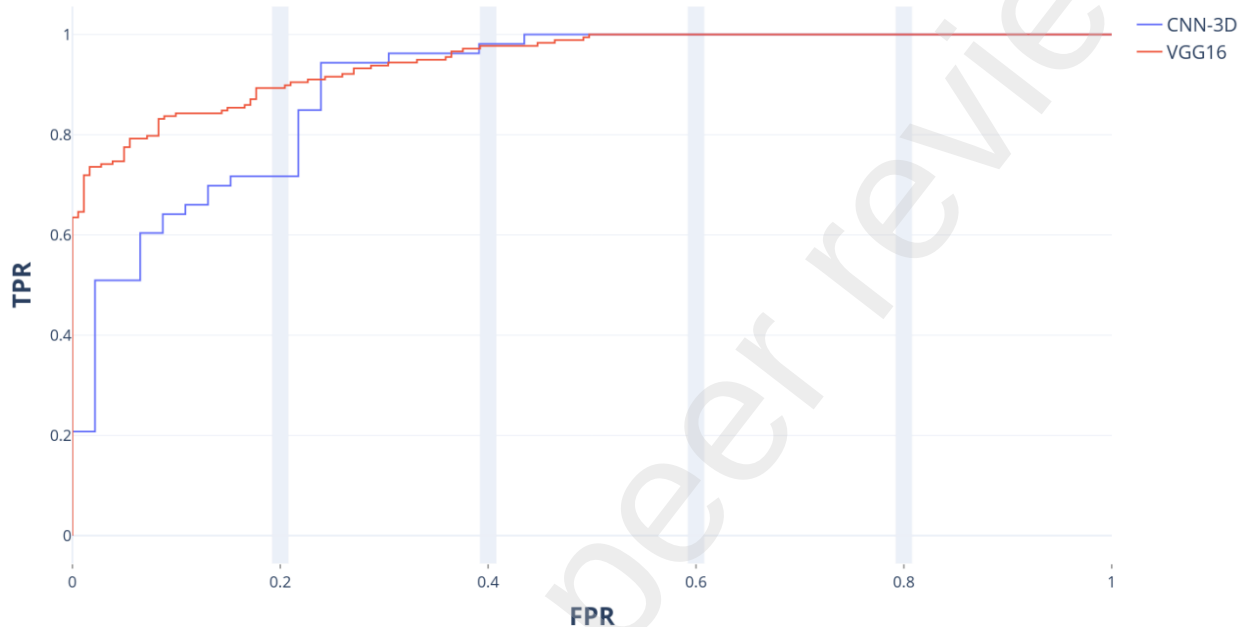


Figure 8. ROC curve, comparing the best performing models between VGG16 and CNN-3D, where TPR is True Positive Rate, and FPR is False Positive Rate.

For the ROI analysis, we focused exclusively on the best-performing VGG16 model. Figure 9 presents the normalized summed weights capturing the change in F1-score from Permutation Feature Importance (PFI) and gradient values from Integrated Gradients (IG) for each ROI, aggregating across all directed interactions involving that ROI. In Figure 9.B, the PFI analysis reveals strong negative scores for the TL, Center, and TR regions, suggesting that shuffling their signals improves model performance. This indicates these ROIs may contribute noise or degrade performance rather than providing meaningful information.

However, when compared with Figure 9.A, which displays IG gradients, the TL and Center regions exhibit near-zero gradient values. This, in conjunction with their negative PFI scores, suggests they are not significant biomarkers for OCD across frequency bands. In contrast, the TR region shows a positive IG gradient despite its negative PFI score. As shown in Figure 9.D, the negative influence of TR persists in its intra-ROI interactions, highlighting its overall detrimental effect. This divergence between PFI and IG often suggests that while the model uses the TR region

for decision-making, its signal is noisy shuffling the input removes false dependencies and thus improves performance. Given the model's strong test metrics, another possible explanation involves heterogeneous contributions from individual channels within the ROI: some channels may drive the positive IG signal, while others introduce noise, reflected in the negative PFI. This hypothesis warrants further investigation in future work.

The PFI analysis also statistically identifies the OR, FR, and FL regions as the most relevant ROIs. From the IG results in Figure 9.A, TR and OL exhibit positive gradients, suggesting that increased connectivity in these regions biases the model toward an OCD classification. In contrast, the OR and FR regions show negative gradients, indicating stronger connectivity here is associated with the healthy (HBN) group. A comparison between Figures 9.A and 9.C reveals that unchanged regions between the two figures have negligible intra-ROI contributions to the model's decisions. Notably, the FR intra-ROI interaction appears significant, with higher FR–FR connectivity associated with healthy individuals. Conversely, the OR–OR interaction lessens the negative gradient for the OR ROI, resulting in a positive net gradient—implying that greater OR–OR connectivity is associated with OCD classification.

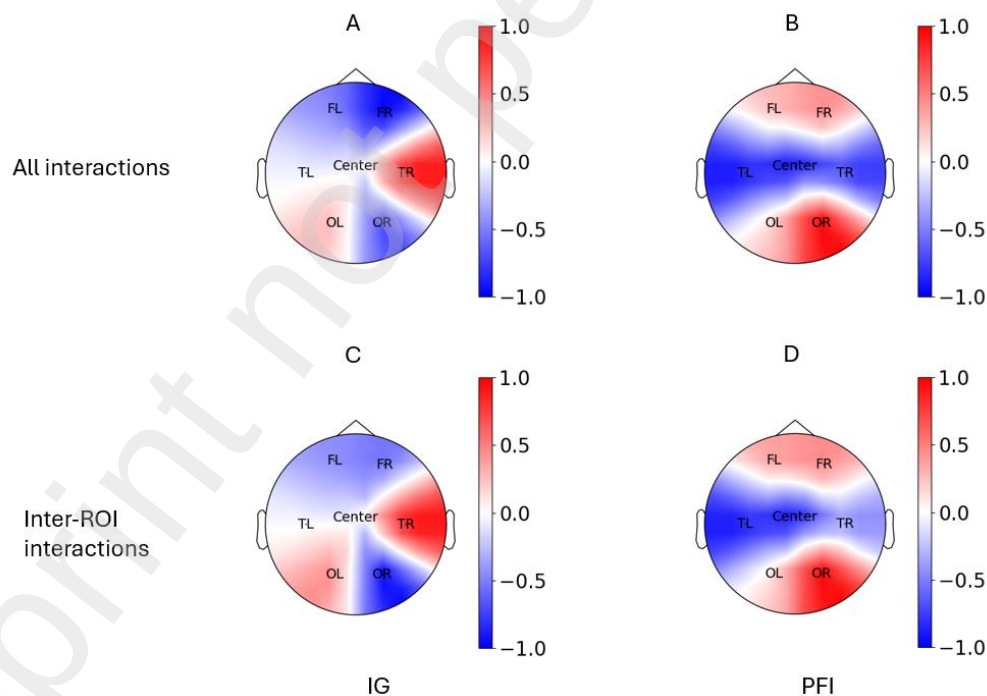


Figure 9. **A.** the normalized IG heatmap for ROI, where all channel interactions were included (inter-, intra- ROI) across frequency bands. **B.** the normalized PFI heatmap for ROI, where all channel interactions were included (inter-, intra- ROI) across frequency bands. **C.** the normalized

IG heatmap for inter-ROI interactions only, across frequency bands. **D.** the normalized PFI heatmap for inter-ROI interactions only, across frequency bands.

In Figure 10, the PFI scores for inter-ROI interactions are shown across different frequency bands. Several specific interactions exhibit negative PFI scores, indicating that they degrade model performance when included. These include TR–Center in the delta band, TR–OL in the theta band, TL–OL and TL–TR in the beta band, TL–TR in the gamma band, and TR–Center across the full combined frequency range.

When examining the overall patterns, we observe that the alpha, theta, beta, and gamma bands contain a greater number of statistically significant inter-ROI interactions, suggesting their higher relevance to the model’s decision-making process. In contrast, the delta band and the general broadband range (0–70 Hz) appear to contribute fewer statistically significant features, implying that inter-ROI interactions in these bands are less informative for the classification task.

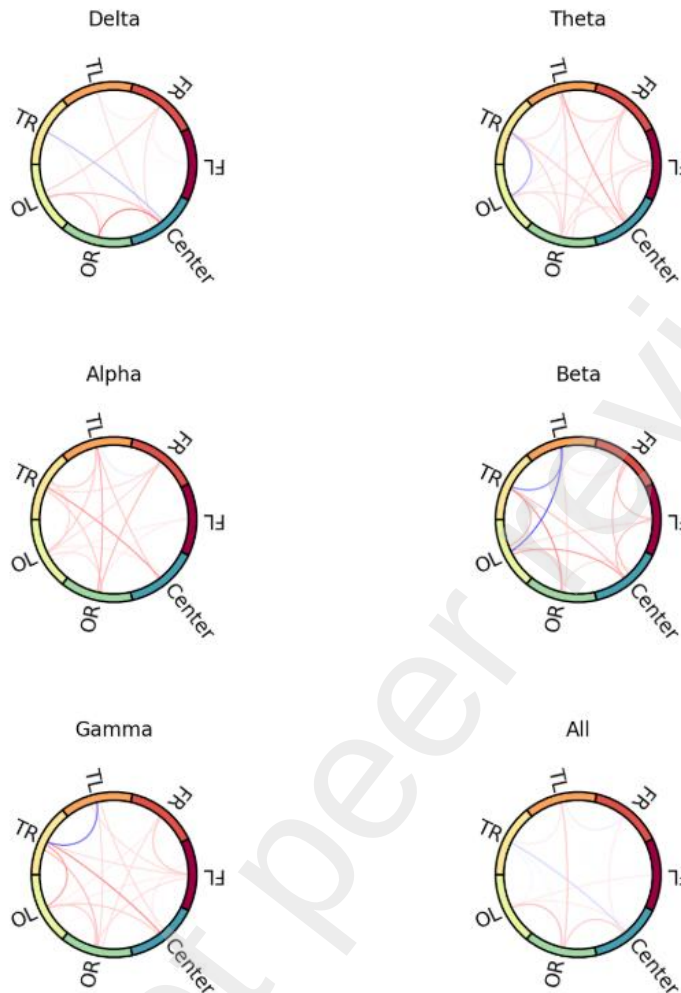


Figure 10. Normalized PFI score for inter-ROI interactions.

Figure 10 presents the gradients derived from the Integrated Gradients (IG) method for inter-ROI interactions across different frequency bands. In the delta band, notable interactions include TR–OL, TR–OR, and TR–Center, which are positively associated with OCD classification. Additionally, TL–FR is highlighted as contributing to the classification of healthy individuals (HBN). However, comparing the TR–Center interaction in the delta band between the PFI and IG results reveals a common discrepancy between these two interpretability methods. Similar inconsistencies are observed in TR–OL (theta), TL–OL (beta), and TR–Center (general frequency band). These patterns suggest that the divergence between PFI and IG scores may be driven primarily by interactions involving TR–TR and its connected regions.

In the gamma band, despite its statistical importance indicated by PFI scores, IG results suggest that its features contribute minimally to the model's decision-making process. A similar

trend is seen in the theta band; however, theta does highlight a single meaningful interaction—
TL–FL—which is associated with HBN classification.

Interestingly, while the general frequency band (0–70 Hz) shows limited statistical
significance based on PFI scores, IG analysis identifies several crucial interactions for model
predictions. These include OL–FL, FL–TL, FR–TL, and FL–OR, all associated with OCD
classification. Conversely, the Center–FL interaction is associated with healthy classification.

Among all frequency bands, alpha and beta exhibit the most influential inter-ROI
interactions. Notably, alpha-band interactions are predominantly linked with OCD classification,
whereas beta-band interactions are more closely associated with HBN classification.

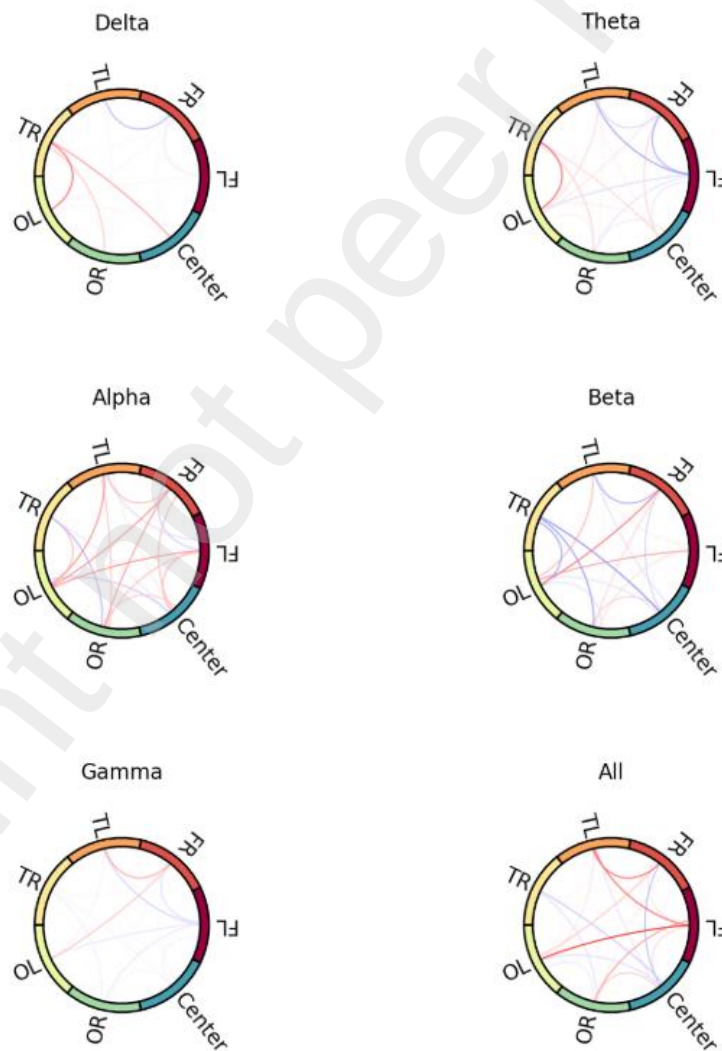


Figure 10. Normalized IG score for inter-ROI interactions.

Table 6. Summary of significant bi-directional fPDC between ROIs, for the IG and PFI methods. The highlighted interactions are ones that show discrepancy between the PFI and IG trends, i.e. statistically noisy with while having a positive or negative gradient in IG method.

Frequency Band	IG		PFI	
	HBN-Biased	OCD-Biased	Statistically Important	Statistically Noisy
Delta	TL-FR	TR-Center , TR-OL, TR-OR	OR-Center, OR-OL, TL-Center, OL-Center, FR-Center, FR-OL	TR-Center
Theta	TL-FL, FL-FR, FR-TL	TR-OL	Majority of interactions	TR-OL
Alpha	OR-TR, OL-Center, FL-TL, FL-FR	Majority of interactions	Majority of interactions	-
Beta	Majority of interactions	OL-FR, OL-FL, OL-TL , OR-Center	Majority of interactions	OL-TL , TL-TR
Gamma	FL-TL, FL-OR, FL-FR	FR-OL, FR-TL	Majority of interactions	TL-TR
All	Majority of interactions	FL-OR, FL-OL, FL-TL, FR-TL, FR-OL	OR-OL, OR-TL, OR-Center	TR-Center

Figure 11 presents the percentage of segmented data classified as OCD for each participant in our test dataset. On average, 76% (± 32) of segments from OCD participants were classified as OCD, while 14% (± 24) of segments from HBN participants were classified as OCD.

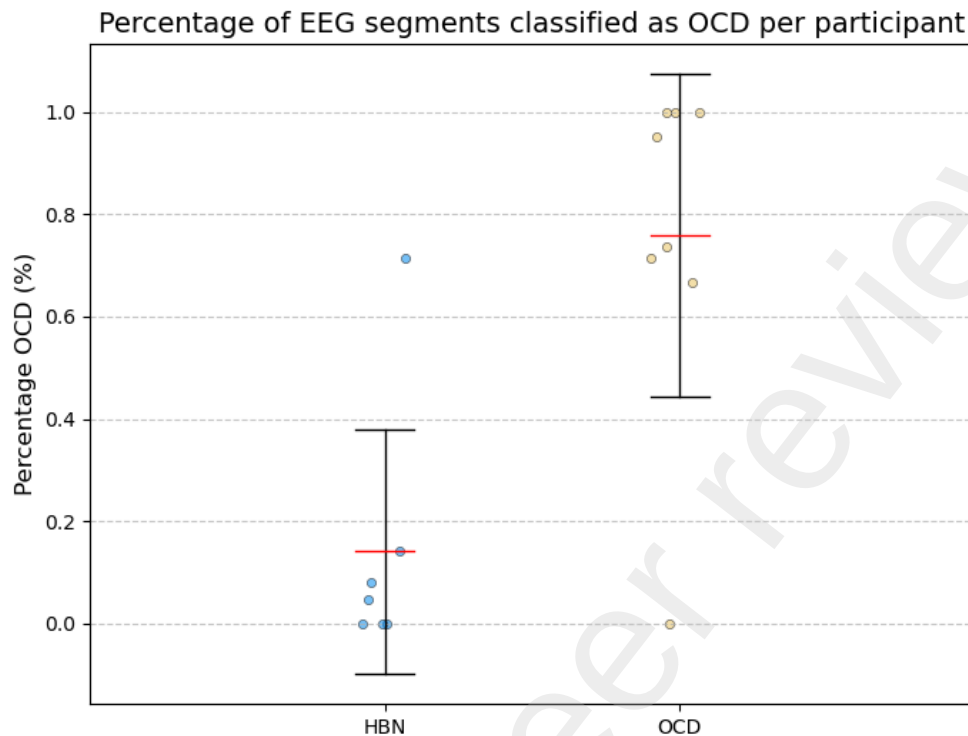


Figure 11. Percentage of participants segments classified as OCD or HBN, for each participant in the test dataset.

Discussion

In this study, we applied CNN models on high-density EEG data recorded from American children and adolescents diagnosed with OCD and healthy control to explore diagnosed based biomarkers. Our results demonstrated that the VGG16 model, when trained on rsFC derived from EEG data, achieved strong performance in classifying children and adolescents with OCD compared to healthy controls. The highest classification performance was achieved using the full-frequency ffPDC measure, yielding an accuracy of 0.866, an F1-score of 0.853, and a high specificity of 0.946. These metrics underscore the model's robust ability to capture neuro-biomarkers associated with OCD. Furthermore, the model's effectiveness was corroborated by additional metrics, including a precision of 0.744, a Matthews Correlation Coefficient of 0.760, and an Area Under the Curve of 0.948, indicating a high level of reliability and diagnostic potential.

In contrast, the CNN-3D model, though promising, underperformed relative to VGG16. It achieved an accuracy of 0.798 and an F1-score of 0.873 using the PDC measure. This lower performance may be partly attributed to the relatively smaller dataset, as 3D convolutional networks typically require larger training sets to generalize effectively. Nonetheless, the CNN-3D

model offers distinct advantages: its ability to process temporal sequences enables it to capture dynamic patterns in functional connectivity, which static models like VGG16 cannot. This temporal sensitivity could provide valuable insights into state-dependent neural fluctuations in OCD, potentially identifying transient disruptions or biomarkers tied to symptom dynamics.

Although VGG16 benefited from access to more data, it is a significantly more complex model than CNN-3D. This complexity was mitigated through transfer learning, which likely contributed to its superior performance. The success of this approach highlights the need for a shared infrastructure for publishing pre-trained EEG models, facilitating transfer learning across research studies. Establishing such a framework would enhance reproducibility, reduce computational overhead, and accelerate the development of EEG-based diagnostic tools. Future research should also prioritize standardizing preprocessing pipelines and model architectures to improve interoperability across studies.

Interpretability analyses revealed a notable discrepancy between the Permutation Feature Importance (PFI) and Integrated Gradients (IG) methods, particularly for the right temporal region. While PFI suggested that perturbing the right temporal region improved model performance implying it may introduce noise, IG revealed positive gradients, indicating its involvement in the classification decision. This inconsistency may stem from heterogeneous channel contributions within the region or from confounding artifacts. Further investigation, potentially at the channel level or using alternative connectivity metrics, is necessary to clarify the right temporal's role.

At the participant level, the VGG16 model correctly classified 76% of segments from OCD participants, while maintaining a low false positive rate of 14% among healthy controls. This segment-level granularity is particularly valuable for clinical applications, as it accounts for the heterogeneity inherent in OCD presentations and may support more individualized diagnostic assessments. The CNN-3D model, with its temporal modeling capabilities, could further enhance these insights by identifying time-dependent patterns or OCD subtypes such as variations in symptom severity or treatment responsiveness.

Our results further highlight distinct patterns in how brain regions communicate across different frequency bands in individuals with OCD compared to healthy controls. Notably, stronger connections in the delta, beta, and gamma bands were more frequently observed in the OCD group, especially in networks involving the right hemisphere (temporal, frontal, and occipital areas). In contrast, connectivity patterns in the theta and gamma with exhibited weaker activities while alpha in general in all regions were higher in OCD compared to healthy control were less clear, with several links showing inconsistency between the analysis methods or appearing statistically noisy. Despite this, there was a strong overall agreement between the IG and PFI methods in identifying key OCD-related differences—though some variations, such as those involving the bilateral temporal and temporal left-Center regions, suggest subtle complexities in how these patterns are interpreted. Overall, these results offer a more refined view of how OCD

affects brain connectivity and point toward the potential for developing EEG-based diagnostic tools that focus on specific frequencies and brain regions.

Interestingly, the TR-Center connection appeared particularly unstable across frequency bands, raising questions about its reliability as a consistent marker. Our results also revealed interesting interactions regarding self-connectivity, where right frontal region self-connectivity or self-regulation was higher in healthy individuals (i.e. low self-regulation in OCD), which was also detected by Ahmari et al. (2022), where the paper discusses how dynamic self-regulatory mechanisms in the prefrontal cortex such as inhibitory control and internal feedback loops are impaired in OCD, which could be interpreted as reduced self-connectivity or self-regularization.

On the other hand, self-connectivity in the right temporal region pops up as a specifically noisy region to the CNN model, while all the other self-connectivity regions are ignored by the model and considered statistically insignificant. As mentioned before, it is possible that this is due to heterogeneous channel contributions within the region or from confounding artifacts. However, there is another possible explanation; in Deco et al. (2011), and Parente et al. (2020), using fMRI, the papers discuss how resting-state networks are dynamically organized and often exhibit anti-correlated patterns. Specifically, frontal executive control regions (like the dorsolateral prefrontal cortex) are often anti-correlated with temporal and parietal regions, especially those involved in the default mode network. In Ahmari et al. (2022), the paper described the low self-regulation of the prefrontal cortex, it also emphasis that the dysfunction in the prefrontal cortex also translate to abnormal interactions with other regions. So, the degrading behavior of the right temporal region can be a direct result of the low self-regulation of the frontal region, which is causing a non-linear relationship with the right temporal right region. Where in some OCD participant, the anti-correlation between the right temporal and frontal region is maintained, but in other OCD participants, this relationship is broken down, causing the self-connectivity of the right temporal region to have a degrading effect on the model's ability to identify OCD biomarkers.

Our results partially align with previous findings in children diagnosed with OCD, as reported by Tan et al. (2022), who observed increased rsFC, particularly in the alpha and gamma frequency bands, using MEG techniques. The elevated alpha-band connectivity was found not only within the well-established cortico-striato-thalamo-cortical circuit (frontal, subcortical midline, and cerebellar regions) but also extended to regions such as the cerebellum and limbic system. In contrast, the increased gamma-band connectivity was primarily confined to the frontal and temporal areas of the brain. Moreover, Safar et al. (2023) also reported increased gamma activity in children diagnosed with OCD compared to healthy controls, particularly in the right temporo-parietal areas, middle temporal gyri, temporal pole, and angular and supramarginal gyri. However, their study focused on brain responses to emotional stimuli. While a direct comparison is not feasible in this case due to the psychological task applied, from a neurophysiological perspective, patients with OCD often experience significant anxiety and depressive symptoms, cognitive and affective dysregulations, which can also impact sensory and perceptual processing, non-verbal

memory and facial processing linked to these areas (see Daros et al., 2014; Lee et al., 2023, Solway et al., 2022; Perera et al., 2023; Zaboski et al., 2021). These dysregulations are associated with anterior cingulate cortex (Cheng et al., 2013), and higher frontal delta and theta activities in OCD patients suffered from cognitive impairments compared to healthy controls (Kamaradova et al., 2016), as well as increased theta activity is linked to neural system vulnerability observed in OCD patients (Perera et al., 2019).

Additionally, the increase in delta and theta activities in (rostral) anterior cingulate cortex in OCD were also attributed to hypofunction of these brain regions contributing to the OCD symptoms compared to healthy control (Velikova et al., 2016); and may reflect ineffective increased response monitoring or compensatory activity in children and adolescents (Suzuki et al., 2022). Cheng et al. (2013) reported interesting findings highlighting altered rsFC involving the posterior cingulate cortex in individuals with OCD, suggesting its relevance to core deficits of the disorder. Their results showed a strong correlation between OCD symptom severity and abnormal connectivity patterns, particularly in regions associated with planning, as well as sensitivity to reward and punishment (Maltby et al., 2005). These findings point to an imbalance within the cingulate-frontal network, which may reflect disruptions in error detection and self-regulation commonly observed in patients with OCD.

Earlier fMRI research (e.g., Gruner et al., 2014; Armstrong et al., 2016) reported that children and adolescents with OCD tend to have stronger connections between the frontal and cingulate regions of the brain. These studies also reported more localized connectivity within the frontal areas, including the supplementary motor and sensorimotor cortices, during rest period. Alpha-band activity is believed to help coordinate brain communication by linking together local areas into broader networks, supporting important cognitive functions like planning, movement, attention, and memory (Klimesch et al., 1999; Palva & Palva, 2011).

Overall, as noted by other researchers in the field (Tan et al., 2022), these findings need to be replicated in larger samples and examined across different age groups throughout the lifespan. While we believe our results using high-density EEG are promising, it's important to acknowledge that, to our knowledge, no prior studies have applied a similar methodological and computational approach in children and adolescents. Therefore, the differences we observed between the patient group and healthy controls—regardless of illness onset, developmental stage, medication status, or symptom severity—should be interpreted with caution. Further research is needed to account for these factors and validate the findings.

Conclusion

Our results point to clear differences in how the brain regions communicate in children and adolescents diagnosed with OCD compared to those of healthy age-matched controls. Specifically, we observed stronger connections in certain brain networks especially in the delta, beta, and gamma frequency bands across frontal, temporal, and occipital regions. These patterns were consistently picked up by both of our analysis methods, which strengthens our confidence in the findings. In addition, higher beta and gamma activities were observed in the fronto-temporal, and occipo-frontal regions. These findings add to the growing body of research on the neurobiological markers of OCD in children and adolescents and highlight the potential of high-density EEG as a practical tool for identifying brain-based markers that could support earlier diagnosis or track treatment progress in research and clinical settings.

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Author contributions statement

A.J.N, KB: Study design, data analysis, writing of the manuscript; **A.J.N, BH:** rewiring-editing. All authors reviewed the manuscript.

Ethic Statement

The data for this study were sourced from the comprehensive medical database “Healthy Brain Network”, which building was approved by the ethics committee of local American Institutions and provided based on Data Usage Agreement (DUA), further ensuring compliance with research guidelines and participant rights. All participants included in this database provided informed consent, agreeing to make their data available for scientific research purposes. This consent aligns with ethical standards and ensures the integrity and confidentiality of participant information throughout the research process.

Data availability statement

All data can be obtained from Dr. Amir Jahanian Najafabadi (amir.jahanian@uni-bielefeld.de) upon reasonable request.

Conflict of Interest

Authors declare no conflict of interest.

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